

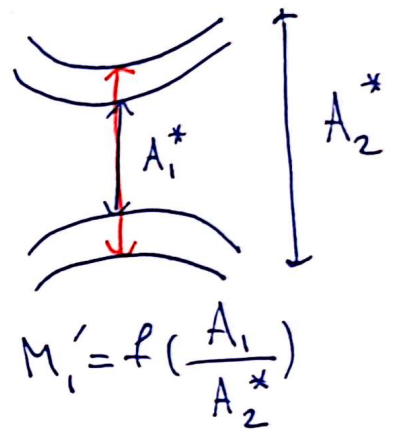
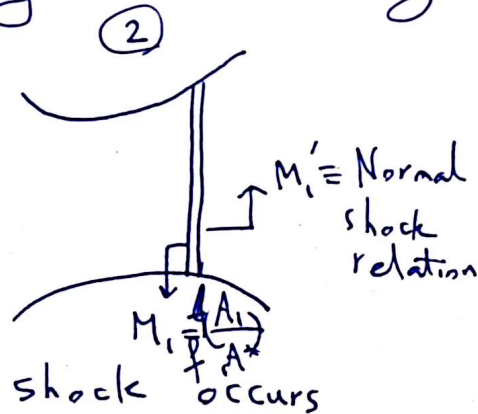
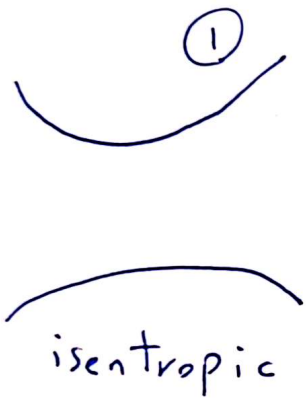
Friday-Feb-21-2025 : Gas Mixture (Section 2)

remember from AE 319.

When a nozzle is isentropic (shock-free) $M_t = 1 \Rightarrow A_t = A^*$

But when there is a shock in nozzle, the flow after the shock does not have the same P_0 as flow before the shock. ($P_{01} \neq P_{02}$).

From m=cte relation across the shock, we can conclude that this new flow belongs to another hypothetical nozzle with A_2^*



Gas Mixture (combustion chamber)

Each constituent of a mixture is called species.

(Air has O_2 species, N_2 species, —)

An ideal gas mixture, at temperature T_m , in a container of volume V_m , has the following characteristics:

$$\underbrace{T_m}_{\text{mixture}} = T_1 = T_2 = \dots = T_n \quad (n: \text{number of species})$$

$$P_m = \sum_{i=1}^n P_i$$

(p-1)

$$V_m = V_1 = V_2 = \dots = V_n$$

$$E_m = \sum_{i=1}^n E_i$$

$$H_m = \sum_{i=1}^n H_i$$

$$S_m = \sum_{i=1}^n S_i$$

" P_i is known as partial pressure"

$$P_i V_m = n_i \bar{R} T_m \rightarrow \text{"number of moles of species } i \text{"}$$

Partial pressure of species i

$$P_1 V_m = n_1 \bar{R} T_m \quad (\text{E.O.S for species 1})$$

$$P_2 V_m = n_2 \bar{R} T_m \quad (\text{ " " " " " 2 })$$

$\vdots$

$$P_n V_m = n_n \bar{R} T_m \quad (\text{ " " " " " } n)$$

Add together: $\underbrace{(P_1 + P_2 + \dots + P_n)}_{P_m} V_m = \underbrace{(n_1 + n_2 + \dots + n_n)}_{n_m} \bar{R} T_m$

number of moles of mixture

$\Rightarrow$ $P_m V_m = n_m \bar{R} T_m$ A mixture of ideal gas behave as an ideal gas

$$P_m \frac{V_m}{m_m} = \frac{n_m}{m_m} \bar{R} T_m$$

note: $m_m = \sum_{i=1}^n m_i$ (mixture mass)

$\downarrow$
 $\frac{1}{\bar{M}_m}$

$$\Rightarrow \frac{P_m}{\rho_m} = \left(\frac{\bar{R}}{\bar{M}_m} \right) T_m = R_m T_m$$

(P-2)

$$R_m = \frac{\bar{R}}{\bar{M}_m} \rightarrow \text{mixture gas constant}$$

$$\Rightarrow \boxed{P_m = \rho_m R_m T_m} \quad \text{E.o.s for mixture}$$

$$\left. \begin{array}{l} \text{Eos for species } i: P_i V_m = n_i \bar{R} T_m \\ \text{" " mixture: } P_m V_m = n_m \bar{R} T_m \end{array} \right\} \text{Divide!}$$

$$\boxed{\frac{P_i}{P_m} = \frac{n_i}{n_m} = X_i}$$

X_i : "mole fraction of species i "

$$\Downarrow \boxed{P_i = X_i P_m} \quad \text{"Dalton's law of partial pressure"}$$

Example: consider air in our room: $\begin{cases} X_{O_2} = 0.2095 \\ X_{N_2} = 0.7808 \end{cases}$

$$P_m = 101 \text{ kPa} \Rightarrow P_{O_2} = X_{O_2} P_m = 0.2095 \times 101 \text{ k} = \boxed{21.23 \text{ kPa}}$$

$$P_{N_2} = X_{N_2} P_m = 0.7808 \times 101 \text{ k} = \boxed{79.11 \text{ kPa}}$$

$$m_m = \sum_1^n m_i = \sum \bar{M}_i n_i, \quad \bar{M}_m = \frac{m_m}{n_m} = \frac{\sum \bar{M}_i n_i}{n_m} = \sum_1^n X_i \bar{M}_i$$

(P-4)³

$$\bar{H}_m = \sum_i^n \bar{H}_i = \sum_i^n n_i \bar{h}_i \quad (\bar{H} = n \bar{h})$$

$$\bar{h}_m = \frac{H_m}{n_m} = \frac{\sum_i n_i \bar{h}_i}{n_m} = \sum_i x_i \bar{h}_i$$

$$\bar{s}_m = \sum x_i \bar{s}_i$$

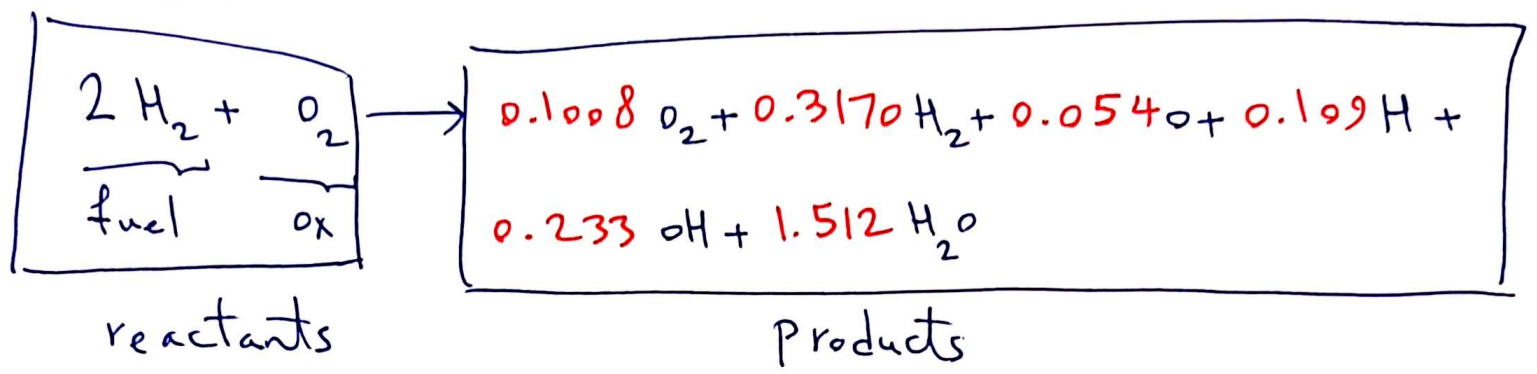
$$\bar{c}_{p,m} = \sum x_i \bar{c}_{p,i}$$

$$\gamma_m = \frac{\bar{c}_{p,m}}{\bar{c}_{v,m}} = \frac{\bar{c}_{p,m}}{\bar{c}_{p,m} - \bar{R}}$$

$$R_m = \frac{\bar{R}}{\bar{M}_m}$$

Example: An ideal gas mixture consists of H_2 & O_2 and has the following properties:

Species ($\bar{M}$)	number of moles (kmol)	$\bar{c}_p$ ($\frac{kJ}{kmol \cdot K}$)	$\bar{h}$ ($\frac{kJ}{kmol}$)	X
O_2 (32)	0.1008	36.3	116233	
H_2 (2)	0.3170	32.9	105346	
O (16)	0.054	20.9	312065	
H (1)	0.109	20.9	283015	
OH (17)	0.233	33.5	149137	
H_2O (18)	1.512	48.27	-87266	



(P-5)